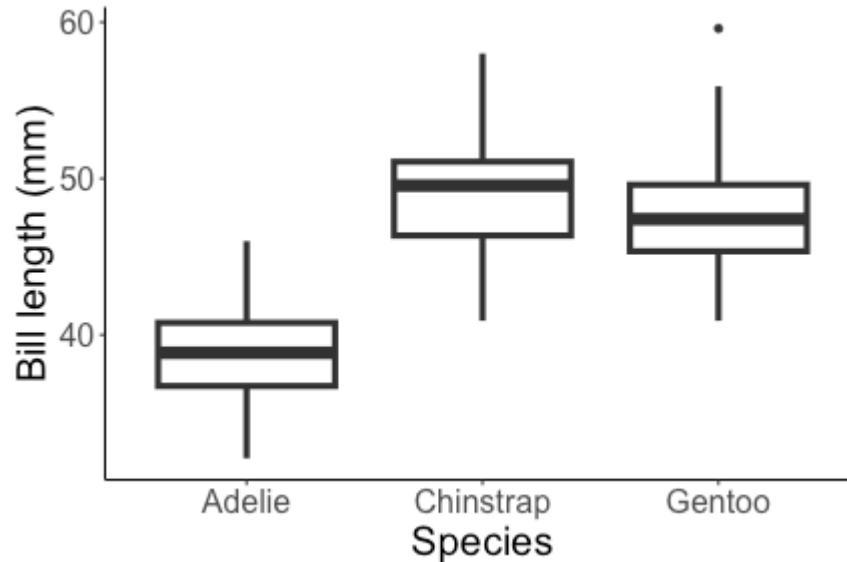


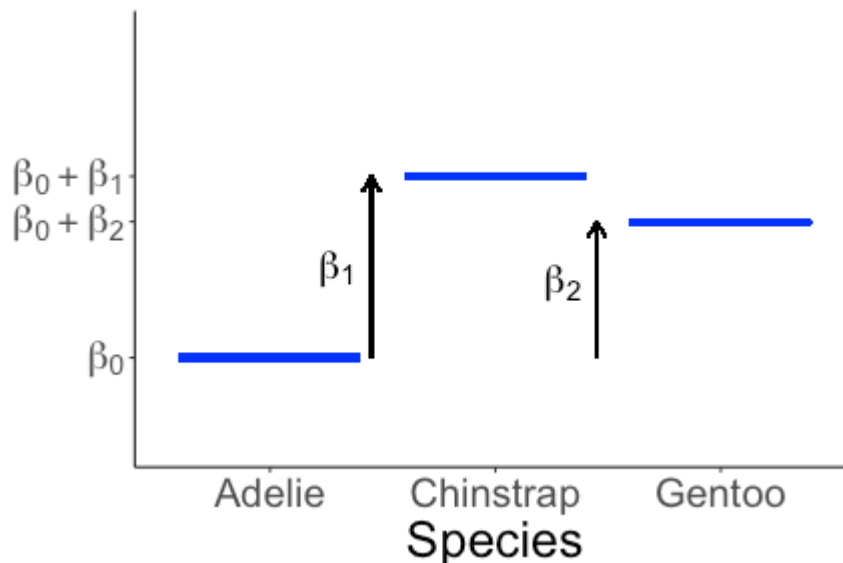
Regression and hypothesis testing with a categorical explanatory variable

Categorical predictor



- + How do we model the relationship between species and bill length?
- + Can we test for a relationship between bill length and species?

Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

How can I write this more concisely?

Indicator variables

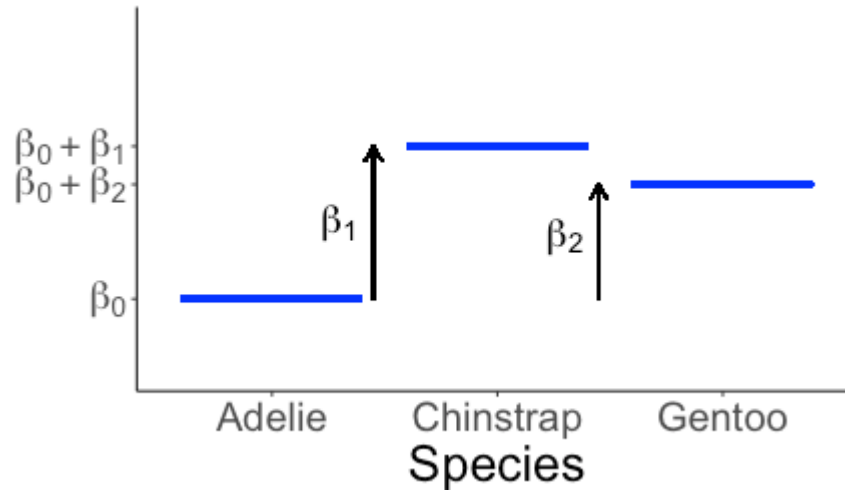
$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Then:

Species	IsChinstrap	IsGentoo
Adelie	0	0
Chinstrap	1	0
Gentoo	0	1

Indicator variables



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Or, more concisely:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Model assumptions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

What model assumptions do you think we make?

Model assumptions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

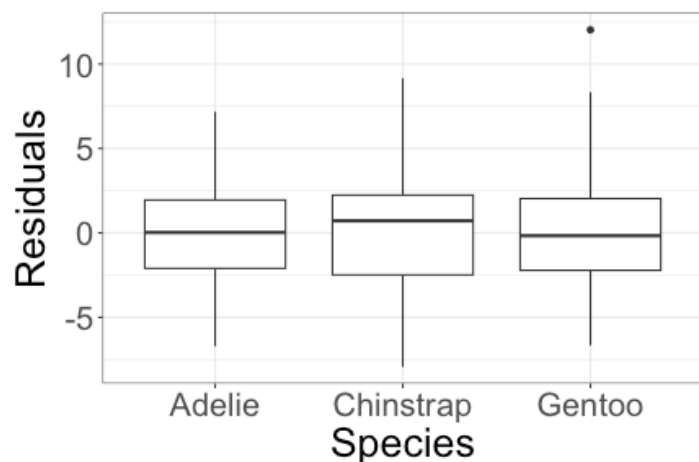
where $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ and the ε are independent of each other.

- + No notion of shape assumption with single categorical predictor
- + Still assume constant variance, normality, independence, and randomness

How should we assess each of these assumptions?

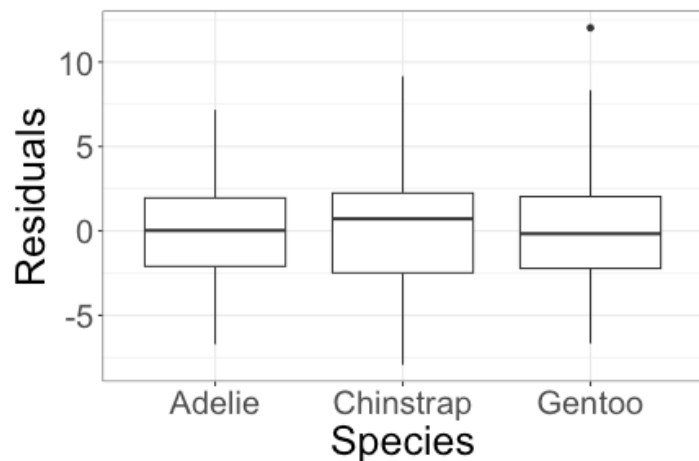
Assessing constant variance

For a single categorical predictor, we can create a boxplot of residuals for each group, and calculate the standard deviation of residuals in each group.



species	residual_std_dev
Adelie	2.66
Chinstrap	3.34
Gentoo	3.11

Assessing constant variance



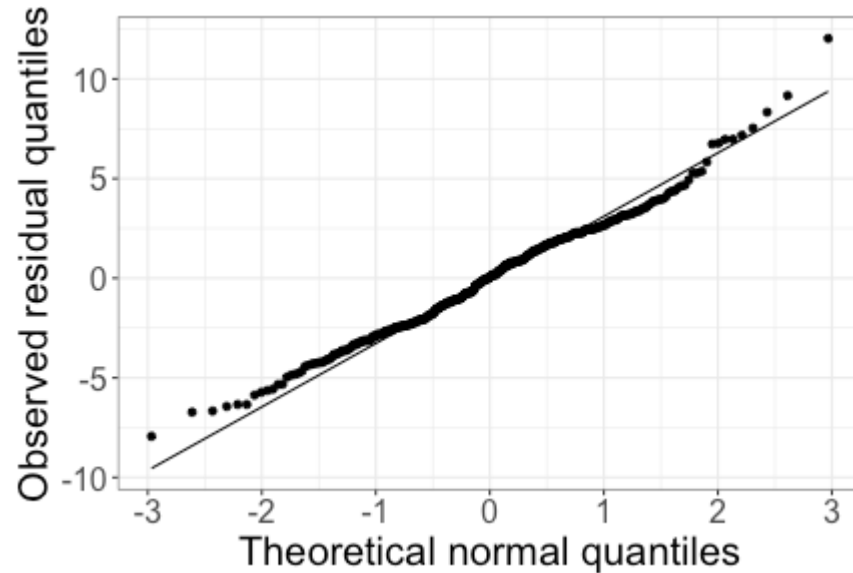
species	residual_std_dev
Adelie	2.66
Chinstrap	3.34
Gentoo	3.11

Looking for similar variability between groups. Rule of thumb: not too concerned about constant variance if

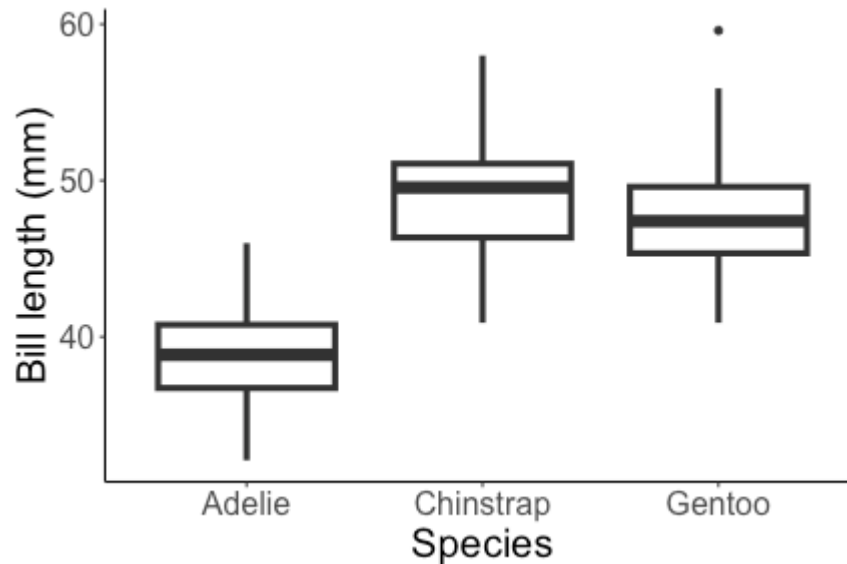
$$\frac{\text{largest standard deviation}}{\text{smallest standard deviation}} < 2$$

Assessing normality

Can use a QQ plot again:



Statistical inference



Does it look like there is a relationship between species and bill length?

Hypothesis testing

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

We want to know whether there is a relationship between species and bill length.

Can I translate this into null and alternative hypotheses about one or more model parameters?

Hypothesis testing

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

We want to know whether there is a relationship between species and bill length.

Can I translate this into null and alternative hypotheses about one or more model parameters?

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Full and reduced models

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

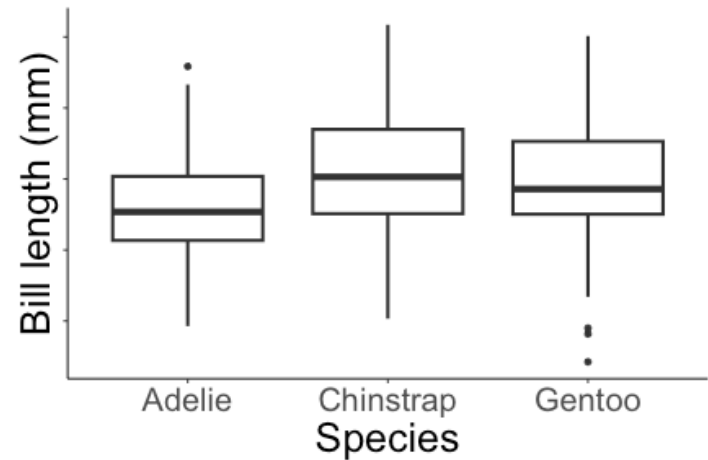
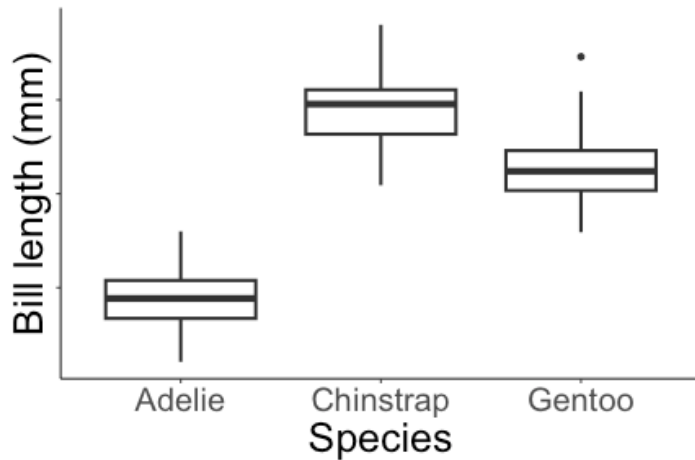
Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

What do I need to test these hypotheses?

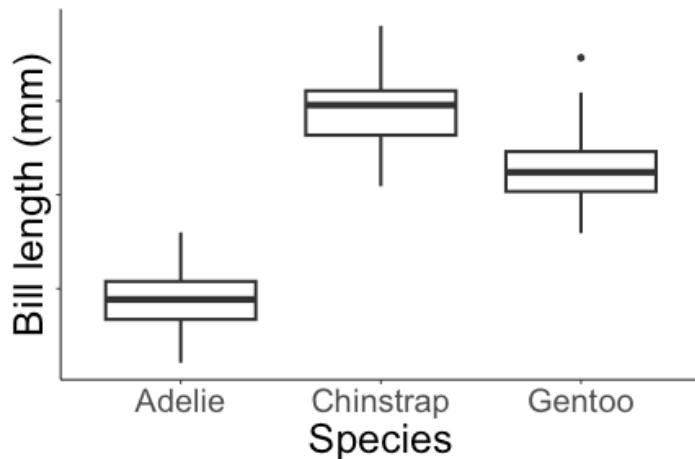
Intuition

Which of these two plots would show more evidence of a relationship between species and bill length?

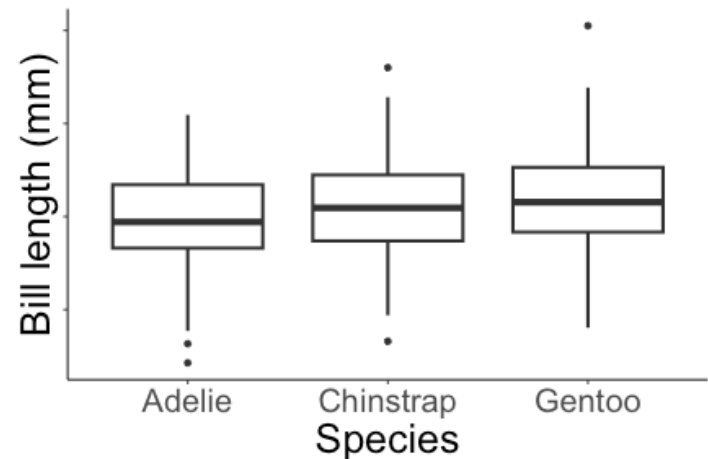


Intuition

More evidence for a relationship when *between-group* variability is larger than *within-group* variability.

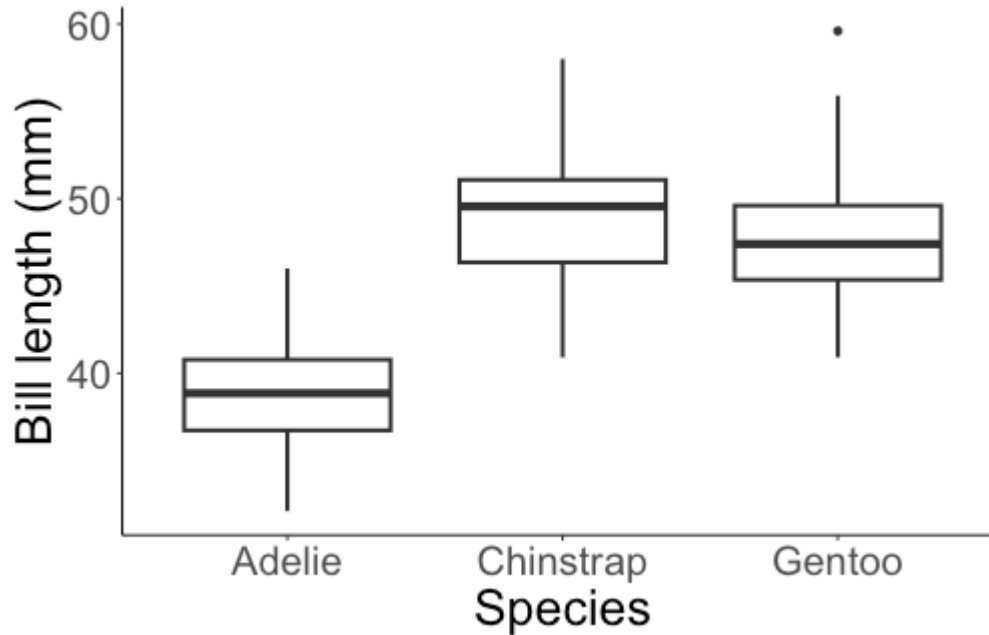


Large between-group variability, relative to within-group variability.



Small between-group variability, relative to within-group variability.

Between-group vs. within-group variability



Between-group variability: $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

Within-group variability: $\sum_{i=1}^n (y_i - \hat{y}_i)^2$

Partitioning variability and degrees of freedom

Variability:

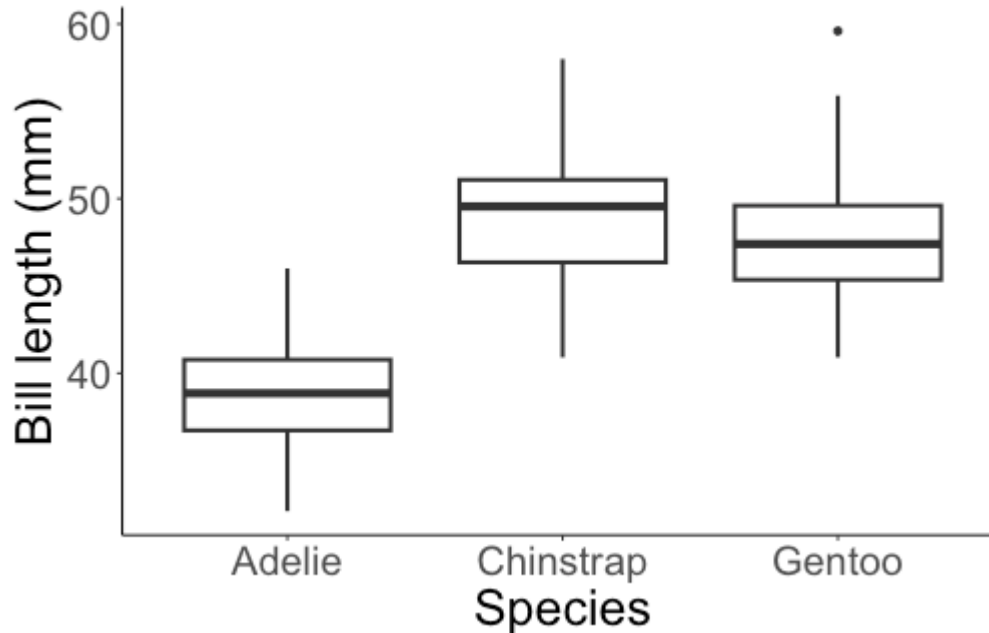
$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSTotal = SSModel + SSE$$

Degrees of freedom:

$$n - 1 = (p - 1) + (n - p)$$

Between-group vs. within-group variability



Test statistic:

$$F = \frac{\text{between group variance}}{\text{within group variance}} = \frac{SSModel/(p - 1)}{SSE/(n - p)}$$

Partitioning variability

Variability:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSTotal = SSModel + SSE$$

Test statistic:

$$F = \frac{\frac{1}{p-1} \sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\frac{1}{n-p} \sum_{i=1}^n (y_i - \hat{y}_i)^2} = \frac{\frac{1}{p-1} SSModel}{\frac{1}{n-p} SSE} = \frac{MSModel}{MSE}$$

Analysis of variance

Test statistic:

$$F = \frac{\frac{1}{p-1} \sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\frac{1}{n-p} \sum_{i=1}^n (y_i - \hat{y}_i)^2} = \frac{\frac{1}{p-1} SS_{Model}}{\frac{1}{n-p} SSE} = \frac{MS_{Model}}{MSE}$$

Analysis of variance (ANOVA) table:

Source	df	SS	MS	F
Model	$p - 1$	SS_{Model}	$MS_{Model} = \frac{SS_{Model}}{p-1}$	$F = \frac{MS_{Model}}{MSE}$
Residual	$n - p$	SSE	$MSE = \frac{SSE}{n-p}$	
Total	$n - 1$	SST_{total}		

Example

$$\sum_{i=1}^n (\widehat{\text{bill length}}_i - \overline{\text{bill length}})^2 = 7015.4$$

$$\sum_{i=1}^n (\text{bill length}_i - \widehat{\text{bill length}}_i)^2 = 2913.5$$

Source	df	SS	MS	F
Model				
Residual				
Total				

Class activity

Work on the class activity (handout).